MIL-O-0050077F(EA)
15 August 1983
SUPERSEDING
MIL-O-0050077E(EA)
3 April 1978
USED IN LIEU OF
MIL-O-50077B
30 September 1964

MILITARY SPECIFICATION

OUTSERT, EYELENS, M1

This limited coordination military specification has been prepared by the US Army Armament, Munitions and Chemical Command based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-O-50077B. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers one type of chemical-biological mask eyelens outsert.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4240

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STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

C5-2-997 - Outsert, Eyelens, M1.
D136-19-14 - Puncture Tester, Natural Rubber, Q38, Assembly.
E136-19-22 - Flexibility Tester, Natural Rubber, Q37, Assembly.

PUBLICATIONS

CHEMICAL RESEARCH AND DEVELOPMENT CENTER DIRECTIVE

No. 618 - Ozone Resistance Test of Natural Rubber Material.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

ASTM STANDARDS

D412 - Tension Testing of Rubber.
D572 - Accelerated Aging of Vulcanized Rubber by the Oxygen-Pressure Method.
D573 - Accelerated Aging of Vulcanized Rubber by the Oven Method.
D624 - Tear Resistance of Vulcanized Rubber.
D2240 - Indentation Hardness of Rubber and Plastic by Means of a Durometer.

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(Application for copies of ASTM Standards should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components. All materials and components listed on Drawing C5-2-997 shall conform to the specifications and drawings forming a part of this specification.

3.1.1 Rubber compound. The rubber compound for the apron shall be non-blooming, except for slight wax-type bloom, and shall contain no less than 65 percent, by volume of new natural rubber. The compound shall be free from lead or mineral sulfates.

3.1.1.1 Sun-checking retardant. The rubber compound shall contain a sun-checking retardant.

3.1.1.2 Copper and manganese inhibitor. The rubber compound shall contain a copper and a manganese inhibitor.

3.1.1.3 Coloring materials. The rubber compound shall be colored by carbon black as required.

3.1.1.4 Test slabs. Test slabs, 6 by 6 inches, by 0.075 $\pm$ 0.01 inch, shall be made from stock representative of the formulation used for the preproduction sample and each lot during regular production of aprons. The slabs shall have a cure equivalent to that of the regular production aprons (see 6.3).

3.1.1.5 Physical properties. Physical properties shall conform to the requirements shown in table I when tested as specified therein. Specimens for tests shall be prepared from test slabs (see 3.1.1.4) as specified in 4.4.4.1.

3.2 Manufacture and assembly. The eyelens outsert shall be manufactured and assembled in accordance with Drawing C5-2-997.

3.3 Preproduction. Prior to the start of regular production, a preproduction sample of eyelens outserts and test slabs shall be produced in accordance with this specification for examination and tests (see 4.3). Any major change in rubber stocks (see 6.4) shall be cause for production of a new preproduction sample.

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TABLE I. Physical properties

Property	Unaged	After heat aging	Test paragraph
Average tensile strength, MPa, min	20.7	17.6	4.4.4.2(a)
Average ultimate elongation, %, min	600	—	4.4.4.2(a)
Percent increase (max)	---	20	---
Percent decrease (max)	---	25	---
Average tensile stress at 200% elongation, MPa	1.72 to 3.10	1.72 to 3.10	4.4.4.2(a)
Individual specimens, MPa, min	1.65	1.65	---
Average tear resistance, kN/m	39.4	28.9	4.4.4.2(a)
Average tension set at 75% ultimate elongation, % max	20	---	4.4.4.2(a)
Average hardness (Shore "A" or "A2")	47 ± 3	---	4.4.4.2(a)
Ozone resistance, min	80	---	4.4.4.2(b)
Flexibility	--	---	4.4.4.2(c)
Units at 24° ± 3°C min	7.5	---	---
Units retained at -32° ± 3°C min	4.8	---	---
*Puncture resistance, psi, min	9.5	---	4.4.4.2(d)

*To be performed on preproduction sample only.

3.4 Workmanship. The apron shall be smooth and free from imperfections, such as pits, bumps, porous areas, cracks, checks, flash, abrasions, or blisters. The rubber shall be compact and free from holes, bubbles, and foreign matter. Ridges or mold parting lines on outer surfaces shall not be considered as flash when closely trimmed. Rips, tears, or punctures shall be considered as damage. The lens shall be free from scratches, cracks, and foreign matter.

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3.5 Resistance to oxygen-pressure aging. The outsert aprons (Dwg C5-2-999) shall show no cracking or increased hardness or tackiness after 72 hours of accelerated aging in oxygen, at a pressure of 2.10 ± 0.10 megapascals and a temperature of 70° ± 1°C, when tested as specified in 4.4.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 15 eyelens outserts, five outsert aprons (Drawing C5-2-999), and 39 test slabs shall be produced using the same methods, materials and equipment as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. Sample eyelens outserts shall be inspected for all requirements of the drawing and this specification.

4.3.2.2 For tests. The test slabs produced in accordance with 4.3.1 (see 3.1.1.4) shall be tested in accordance with 4.4.4. The outsert aprons shall be tested in accordance with 4.4.4.3.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified by 4.3.2.1 and 4.3.2.2 to be acceptable. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

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4.4 Quality conformance inspection.

4.4.1 Lotting. A regular production lot shall consist of the eyelens outserts produced by one manufacturer from the same materials, at the same plant, and under essentially the same manufacturing conditions. No lot of eyelens outserts shall contain more than one lot of aprons produced from one batch of rubber (see 6.6) or more than one lot of plastic lenses.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Thirty-nine test slabs produced in accordance with 3.1.1.4 shall be taken for physical property tests. Ten outsert aprons shall be taken at random from each lot for testing resistance to oxygen-pressure aging.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample eyelens outserts from the regular production lots shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive tests. Sample test slabs shall be tested in accordance with 4.4.4. The acceptance number is zero. The outsert aprons shall be tested in accordance with 4.4.4.3. The acceptance number is zero.

4.4.3.3 Classification of defects.

(a) Outsert, eyelens, M1 (Dwg C5-2-997).

Categories		Defects
Critical:		None defined
Major:	AQL 2.5 percent defective	
101		Component missing or incorrectly assembled
102		Workmanship (see 3.4)
Minor:	AQL 6.5 percent defective	
201		Instruction card missing or incorrect, when applicable
202		Marking (instruction card or pouch) illegible or incorrect
203		Rubber thickness incorrect
204		Color incorrect

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4.4.4 Tests.

4.4.4.1 Rubber test specimens. Specimens shall not be buffed or stretched before testing. Twenty specimens, two from each of 10 slabs, shall be taken for tensile strength, ultimate elongation, and tensile stress testing. Ten specimens shall be tested before aging and 10 tested after aging. Six specimens, two from each of three test slabs, shall be taken, without aging, for hardness and tension set testing. Twenty specimens, two from each of 10 test slabs, shall be taken for tear resistance testing. Ten specimens shall be tested before aging and 10 tested after aging. Twelve specimens, two from each of six test slabs, shall be taken for ozone resistance testing. Eight specimens, two from each of four test slabs, shall be taken for flexibility testing. Six specimens, one from each of six test slabs, shall be taken for puncture resistance testing. The test specimens selected for heat aging shall be aged in an air oven at a temperature of 70 ± 1°C for a continuous period of 168 ± 1 hours or at a temperature of 100 ± 1°C for a continuous period of 24 ± 1 hours, as specified in ASTM D573 (see 6.7).

4.4.4.2 Rubber physical properties tests.

(a) Tensile strength, ultimate elongation, tensile stress, tear resistance, tension set, and hardness. Tensile strength, ultimate elongation, tensile stress, and tension set shall be determined in accordance with ASTM D412. Tear resistance shall be determined in accordance with ASTM D624. Hardness shall be determined in accordance with ASTM D2240. The angle-type die shall be used for tear resistance.

(b) Ozone resistance. Ozone resistance shall be determined in accordance with Directive No. 618.

(c) Flexibility. Flexibility shall be determined with Q37 Natural Rubber Flexibility Tester Assembly (D136-19-14). Flexibility shall be read on the grid of the apparatus. Four specimens shall be tested at 24 ± 3°C and four specimens shall be tested at -32 ± 3°C after being conditioned for 72 hours at that temperature.

(d) Puncture resistance. Resistance to hydrostatic pressure shall be determined with Q38 Natural Rubber Puncture Tester Assembly (E136-19-22).

4.4.4.3 Resistance to oxygen-pressure aging. The outsert aprons shall be tested in accordance with ASTM D572.

5. PACKAGING

5.1 Unit packing, packing, and marking, interplant shipment (see 6.8). The outsert eyelens shall be unit packed and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with Fed. Std. No. 123 and shall include the contractor's name, address, and contract number.

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5.2 Repair parts. When outserts, eyelens are procured for storage and issue as repair parts, preservation, unit packing, packing and marking shall be as specified on the packaging data sheet which is identified by the National Stock Number.

6. NOTES

6.1 Intended use. The eyelens outsert is intended for use with the M17, M17A1 and M17A2 Masks to prevent frosting of eyelens under arctic conditions.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Equivalent cure. Slabs and aprons of the same stock and gage are considered to have equivalent cures or the same state of vulcanization when:

(a) Their free-sulfur contents do not vary more than 10 percent.

(b) Their average tensile-stress values at any given elongation (other than ultimate elongation) do not vary more than $\pm$ 2.5 percent.

(c) Their curing times can be equated in the following equation:

$$t_2 = t_1 \times 1.6^{(T_1 - T_2)/10}$$

where: t_1 = minutes of cure at temperature T_1,

t_2 = minutes of cure at temperature T_2,

T_1 = temperature °F of cure for time t_1, and

T_2 = temperature °F of cure for time t_2.

For convenience, the nomograph in figure 1 may be used instead of the mathematical computation of time-temperature relationships in determining equivalent cure.

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EQUIVALENT CURE TIME FOR NATURAL RUBBER

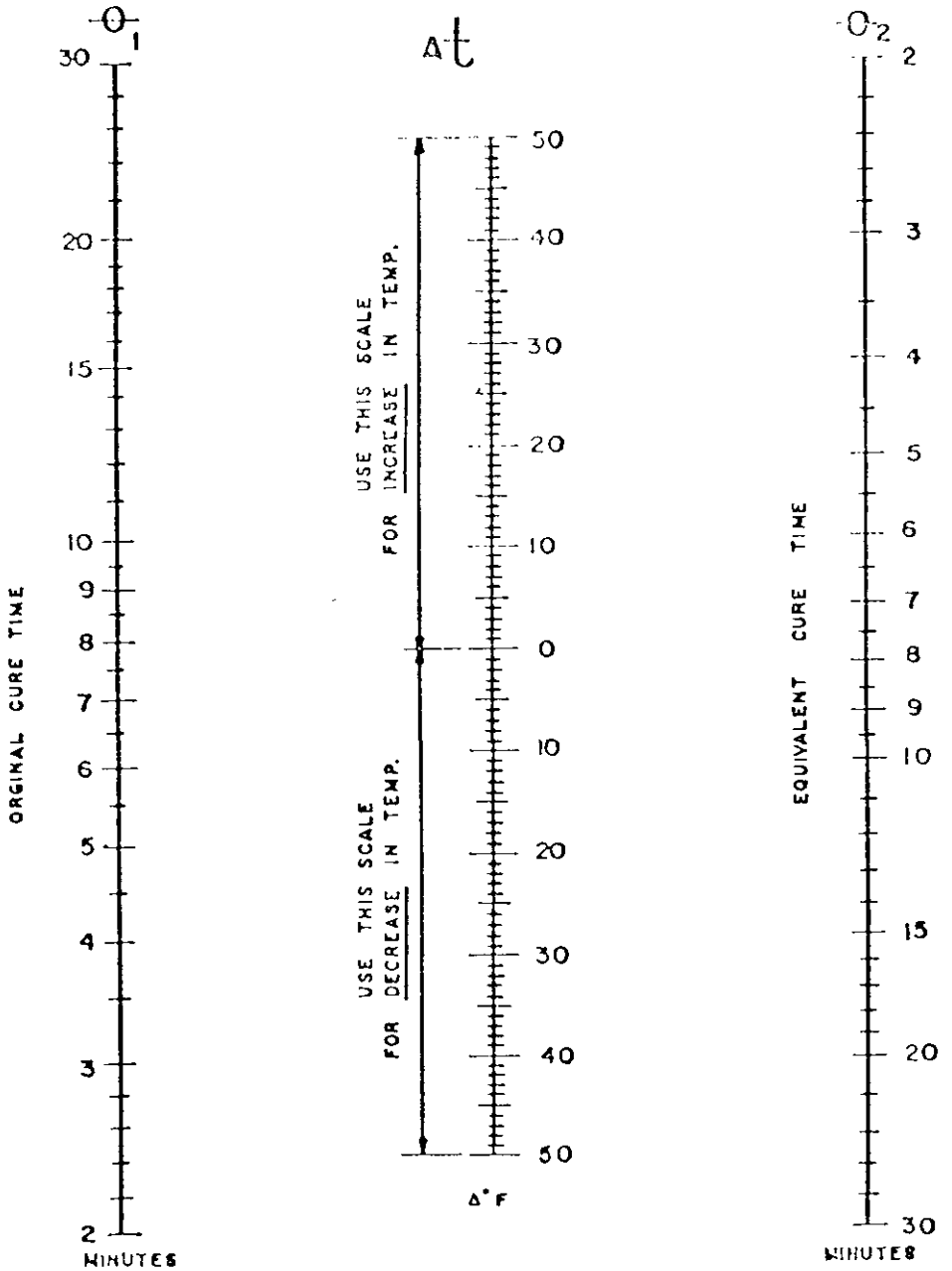


Figure 1.

USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS θ_2 SCALE AT THE EQUIVALENT CURE TIME.

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6.4 Major change. The term applies to any change in formulae, processing, or curing of currently approved rubber stocks. The major change should be considered effected when one or more of the followng conditions occur:

(a) Components of formulae are varied in excess of the limitations stated in the manufacturer's tolerances (see 6.5).

(b) The addition or deletion of one or more components of formulae.

(c) Substitutes for one or more components or formulae without advance approval of Chemical Research and Development Center.

(d) Deviation from the submitted and approved factory mold cure temperature exceeding ± 5°F, even though equivalent cure is attained.

(e) Use of mold lubricants, dusting powders, or wet sprays not used in preparation or cure of the original samples submitted for approval.

6.5 Manufacturer's tolerances. The manufacturer should submit to the contracting officer the variations in amounts used, or tolerances, for accelerators, plasticizers, and softeners. These tolerances are permitted to enable the manufacturer to produce stock which will consistently comply with this specification and will process satisfactorily in the plant. Components of formulae may be submitted on the basis of percentage of stock or number of parts of rubber, and the manufacturer's tolerances recorded in the same units.

6.6 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.7 Aging. Aging at 100 ± 1°C for a continuous period of 24 hours is slightly more severe on rubber than aging at 70° ± 1°C for 168 consecutive hours; therefore, specimens that fail to meet requirements when aged at the higher temperature may be discarded and replaced by new specimens aged at 70 ± 1°C for 168 consecutive hours.

6.8 Interplant shipments. Packaging for supplies and materials which will not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

Custodians:

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Preparing activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-0050077F(EA)	OUTSERT, EYELENS, M1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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